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The Editor-in-Chief
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Subject: Submission of an original research manuscript — "A Sustainable Hybrid QKD–PQC Security Architecture for Quantum-Resilient Residential IoT"

Dear Editor,

On behalf of my co-authors and myself, I am pleased to submit the original research manuscript entitled "A Sustainable Hybrid QKD–PQC Security Architecture for Quantum-Resilient Residential IoT" for consideration in the *IEEE Internet of Things Journal*. The manuscript has not been published previously and is not under consideration for publication elsewhere.

Scope and Significance

Battery-powered residential IoT nodes must remain operational for years while facing the emerging threat of quantum cryptanalysis and the "harvest-now, decrypt-later" attack model. Migrating conventional public-key cryptography to standardized post-quantum cryptography (PQC) directly on constrained leaf nodes, however, imposes a severe energy and radio-bandwidth penalty because large lattice-based keys are split across many IEEE 802.15.4/6LoWPAN frames. This paper addresses that sustainability tension with a gateway-centric hybrid architecture that keeps asymmetric post-quantum operations off the leaf nodes while preserving long-term post-quantum resilience. The work fits the Journal's scope at the intersection of IoT security, edge/energy sustainability, and next-generation network protocols.

Key Points and Contributions

1. **Sustainable hybrid QKD–PQC architecture.** An edge-centric Quantum-Safe Home Security Gateway (QHSG) structurally isolates security-intensive key-management operations from high-volume application data streams; last-mile Quantum Key Distribution (QKD) material is ingested where optical infrastructure exists, while standardized ML-KEM-768 (FIPS 203) provides a deployable path over conventional networks. Bulk data is protected by authenticated symmetric encryption.

2. **Grounded security–sustainability model.** Endpoint compute energy, radio transmission overhead, and 6LoWPAN fragmentation penalties are formalized as explicit design parameters. Constants are anchored to the IEEE 802.15.4 physical layer and the TI CC2652R

datasheet (P_{TX}=22 mW, P_{RX}=21 mW at 250 kbps), with compute times from measured ARM Cortex-M4 cycle counts.

3. **Quantified energy outcome.** Direct leaf-node PQC deployment raises endpoint key-negotiation energy by +108.4% over classical handshakes (328.46 mJ vs. 157.63 mJ at N=100) because the 2272-byte ML-KEM-768 handshake fragments into 24 link-layer frames. The proposed gateway-mediated hybrid reduces endpoint key-management energy by −96.8% relative to standalone PQC and −93.3% relative to classical ECDH.

4. **Protocol-level validation.** The fragmentation model is validated in *executed* headless Contiki-NG/COOJA runs across seven payload sizes (64–2272 B); the measured frame counts (1, 2, 3, 6, 11, 13, 24) match the closed form $\lceil(\text{payload} + 12)/96\rceil$ exactly.

5. **Formal security verification.** A suite of ProVerif 2.05 models machine-checks key-lifecycle secrecy and authentication, single-use refresh semantics, endpoint isolation under device compromise, gateway-compromise forward secrecy, harvest-now-decrypt-later resistance, cross-session/device key independence, ticket replay under packet loss, and active local-link impersonation.

6. **Reproducibility.** Every quantitative and symbolic result is regenerable from a public, tagged artifact containing one run script per evidence stream, including the Contiki-NG patch required to transmit the 2272-byte handshake.

Data and Reproducibility

All simulation code, grounded parameter sets, figure-generation routines, sensitivity-analysis scripts, ProVerif models, cycle-calibration scripts, and Contiki-NG/COOJA scenarios are openly archived at:

https://github.com/prakashsurywanshi/quantum-safe-residential-iot
(branch `ieee-submission`, tag `ieee-v1.0`)

so that reviewers can independently reproduce each claim.

We believe this work offers a rigorous, hardware-grounded, and formally verified framework for sustainable quantum-resilient IoT security, and we hope it is found suitable for publication in the *IEEE Internet of Things Journal*. We would be glad to provide any additional information or artifacts requested.

Yours sincerely,

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